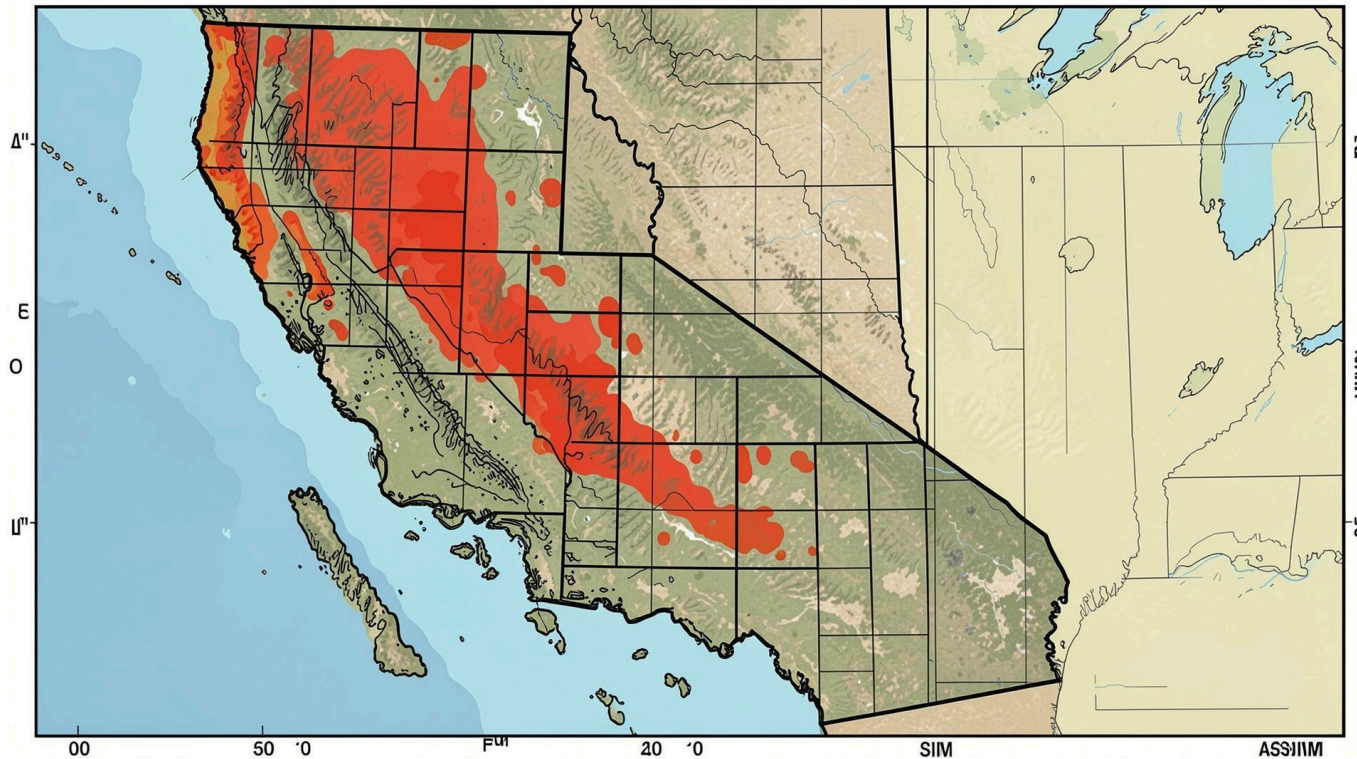


AI-Powered Wildfire Early Detection

Milestone 4 · Numerical Next-Day Wildfire Forecasting · Group 8 · IITM DS & AI Lab



Problem → M4 Goal



Prediction Target

Predict **next-day fire risk** (y_{fire} on $D+1$) for every California **0.25° cell** (~672 land cells).

Realtime Contract

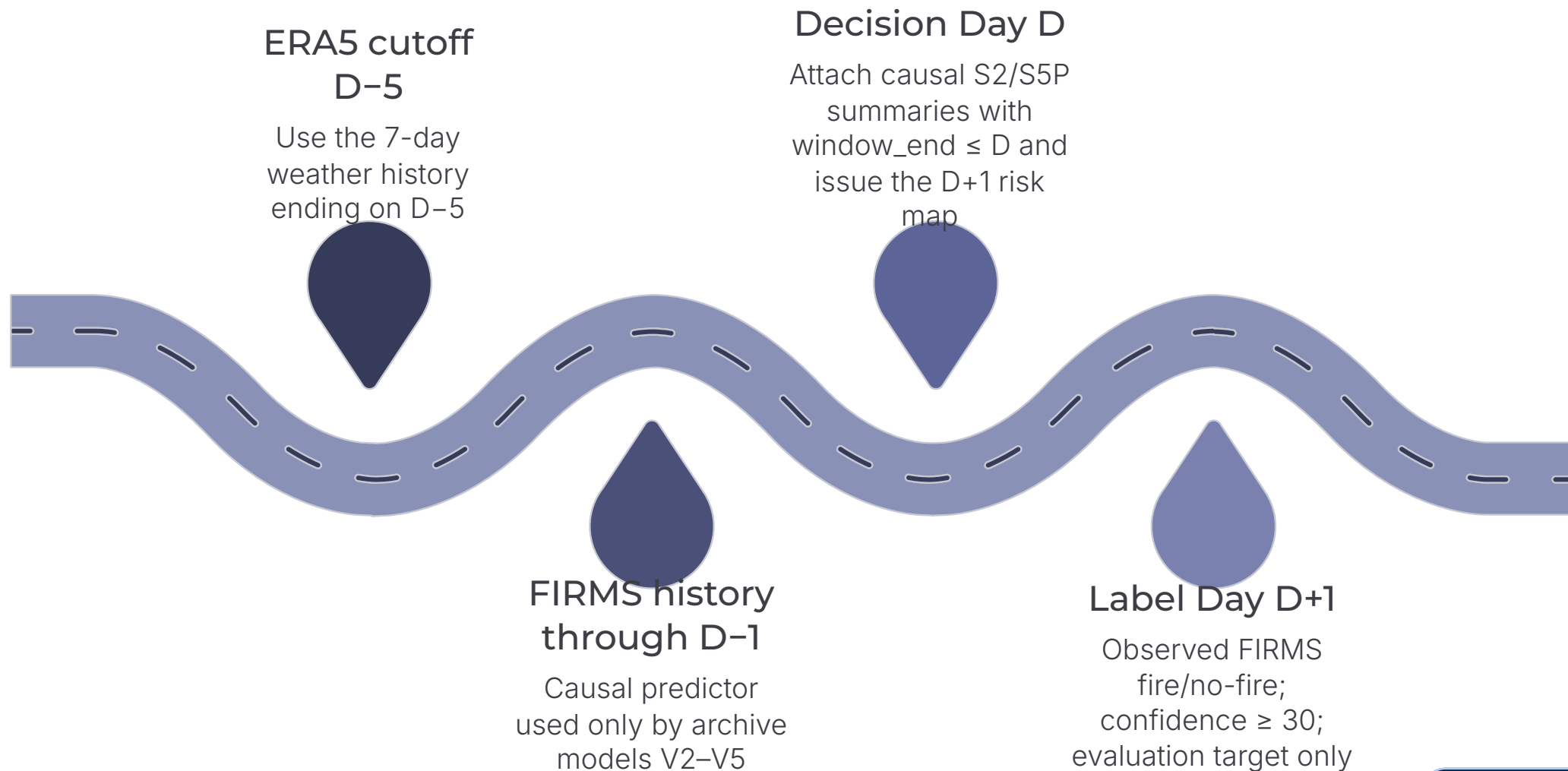
- ERA5 predictors end at **D-5** (publish delay)
- EO (S2/S5P) only if available by decision day **D** — causal

Deliverables

Trained models + HP tuning + PR/ROC + **Recall@K** + **365 daily risk maps**

Two complementary numerical tracks — same labels/split, different feature recipes.

Forecast Contract



Two Numerical Tracks

Same D+1 / D-5 task & chronological split — **fair comparisons only when feature sets match.**

Dimension	Stage A/B/C Pipeline	Archive V1-V5
Starting point	GCS / M3 build → Stage tables	Prepared lag-5 archive tables
Focus	Progressive modalities + LGBM/MLP + month router + maps	Causal fire-history features + ranking + Recall@25
Extra predictors	No FIRMS-as-feature	V2-V5: FIRMS history through D-1
Best model without FIRMS-history predictors	Stage C MLP — PR-AUC 0.0865	V1 classifier — PR-AUC 0.0764
Best model in each track	Stage C MLP — PR-AUC 0.0865	V4 classifier — fire-season PR-AUC 0.2167

V4 uses causal FIRMS history through D-1. Its comparison with the Stage C MLP is a best-system comparison, not a pure architecture comparison.

Data & Stage A → B → C

Feature Progression

Stage A — ~33 features

ERA5 + 7d rollups + DEM

Stage B — ~53 features

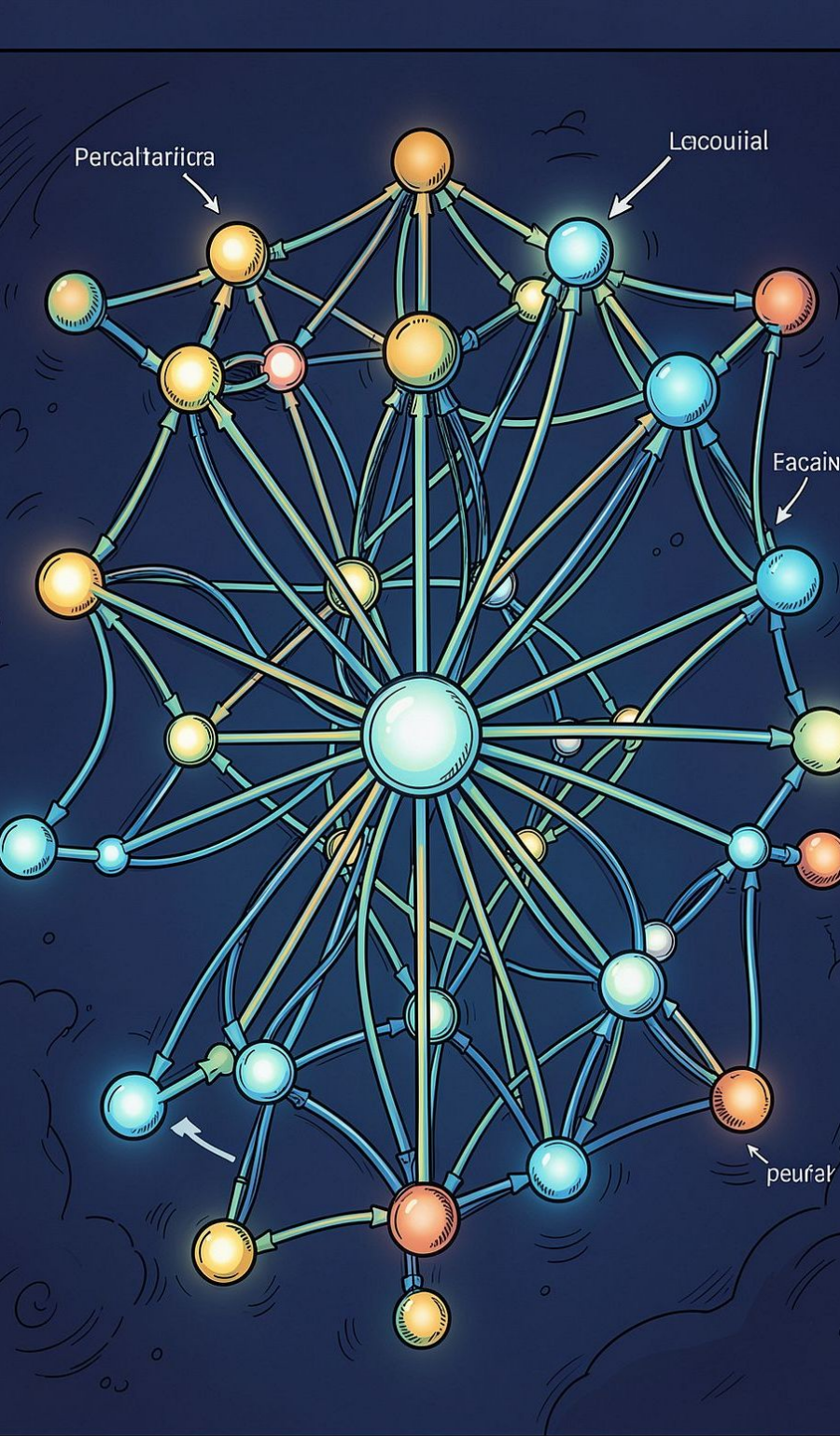
A + Sentinel-2 numerical

Stage C — ~63 features

B + Sentinel-5P numerical

Chronological Split

Split	Years	Rows	Positives
Train	2019–2022	981,792	13,804
Val	2023–2024	491,232	5,536
Test	2025	245,280	2,275



Track A Architecture



Progressive Tables

LightGBM (+ isotonic) primary · **MLP** secondary



Month Router

fire_season (Apr–Nov) + jan/feb/mar/dec



Loss & Selection

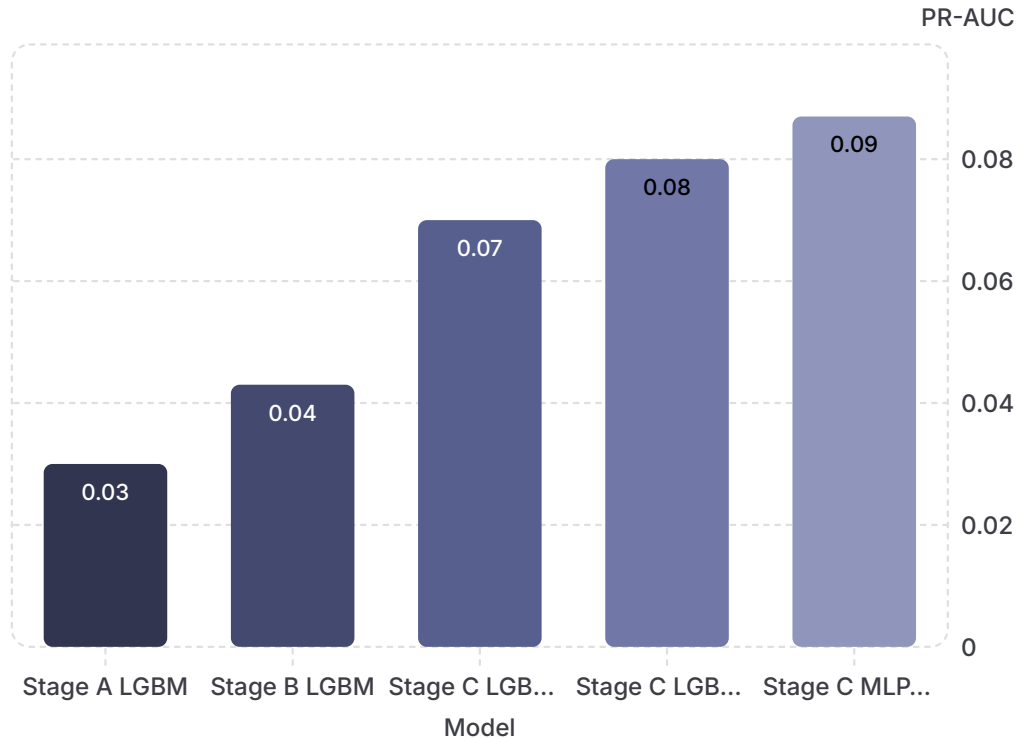
Binary log-loss / BCE + class weights · select on **val PR-AUC**



Leakage Controls

Temporal split · lag identity · causal EO · train-only medians

Track A Results — Fire Season Test 2025



A = Stage A LGBM, B = Stage B LGBM, C = Stage C default, C-L31 = Stage C leaves-31, C-MLP = Stage C MLP

Key Findings

- Clear **Stage A → B → C** improvement under the **D-5 ERA5 constraint**.
- Stage C MLP reaches approximately **8.7×** the chance PR-AUC.
- Validation-selected LightGBM setting: **learning_rate = 0.03**.
- July PR-AUC is approximately **0.208**.
- Lag-0 LightGBM PR-AUC **0.0797** is almost identical to the best lag-5 LightGBM PR-AUC **0.0796**, showing that weather delay is a constraint but not the only bottleneck.

□ Chance baseline ≈ 0.01 (prevalence). Stage C MLP PR-AUC: **0.087**, ROC-AUC: 0.781.

Track B Ladder — Archive V1–V5

Version	Idea	Role
V1	Stage A/B/C LGBM + MLP baselines	Archive baseline
V2	Lagged weather + fire recurrence + neighbourhood history	Largest jump
V3	Direction / distance spread features · mixture · LambdaRank tests	Feature + arch search
V4	Optimise daily Recall@25 · classifier + LambdaRank blend	Recommended
V5	Rerank V4 top set with hard negatives	Extra complexity, little gain

📄 V2–V5 require FIRMS through **D-1** at serve time.

Recommended Model: V4

V4 = regularised LightGBM classifier + daily LambdaRank · 50/50 percentile blend

Full-year 2025 descriptive test metrics

Metric	Value
Calibrated PR-AUC	0.1751
ROC-AUC	0.8420
Brier score	0.00831
Precision@25/day	9.56%
Recall@25/day	38.33%
Recall@50/day	48.88%

 V4 is recommended only when FIRMS history through D-1 is available at serving time.

These metrics are descriptive because 2025 was inspected during iterative experimentation; V4 is not yet production-validated.

What We Learned & What's Next

Delay is a constraint, not the only one

Lag-0 oracle $\approx$ lag-5 on PR — other signal gaps limit performance.

Fire history matters

D-1 FIRMS history drives the largest PR-AUC jump (V1 $\rightarrow$ V2).

Ranking blend provides marginal Recall@25 lift

During walk-forward selection, classifier-only mean Recall@25 was 42.84% versus 42.85% for the 50/50 blend. Recall@50 changed from 55.05% to 55.00%. The blend is retained for V4, but the gain is very small.

Calibrate probabilities and choose the operational alert budget K using the cost of missed fires and false alerts.

Verify real-world availability of D-1 FIRMS before deployment.

Freeze the architecture and operating policy.

Evaluate on a fresh 2026 season or external geography.

Monitor PR-AUC, Recall@ K , calibration drift, and false-alert burden.

